

Exercise 1:

```
Welcome to Cloud Shell! Type "help" to get started.
Your Cloud Platform project in this session is set to elevated-apex-436608-d3.
Use "gcloud config set project [PROJECT_ID]" to change to a different project.
zhuldizkuatkizy@cloudshell:~ (elevated-apex-436608-d3)$ pip install Flask
DEPRECATION: Python 2.7 reached the end of its life on January 1st, 2020. Please upgrade your Python as Python 2.7 is no longer supported. More details about Python 2 support in pip can be found at https://pip.pypa.io/en/latest/development/release-procedures/#python2-support.
Defaulting to user installation because normal site-packages is not writeable
Collecting Flask
  Downloading Flask-1.1.4-py2.py3-none-any.whl (94 kB)
    |#####| 94 kB 1.3 MB/s
Collecting itsdangerous<2.0,>=0.24
  Downloading itsdangerous-1.1.0-py2.py3-none-any.whl (16 kB)
Collecting Jinja2<3.0,>=2.10.1
  Downloading Jinja2-2.11.3-py2.py3-none-any.whl (125 kB)
    |#####| 125 kB 9.7 MB/s
Collecting Werkzeug<2.0,>=0.15
  Downloading Werkzeug-1.0.1-py2.py3-none-any.whl (298 kB)
    |#####| 298 kB 30.1 MB/s
Collecting click<8.0,>=5.1
  Downloading click-7.1.2-py2.py3-none-any.whl (82 kB)
    |#####| 82 kB 950 kB/s
Collecting MarkupSafe>=0.23
  Downloading MarkupSafe-1.1.1-cp27-cp27mu-manylinux1_x86_64.whl (24 kB)
Installing collected packages: itsdangerous, MarkupSafe, Jinja2, Werkzeug, click, Flask
WARNING: The script flask is installed in '/home/zhuldizkuatkizy/.local/bin' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed Flask-1.1.4 Jinja2-2.11.3 MarkupSafe-1.1.1 Werkzeug-1.0.1 click-7.1.2 itsdangerous-1.1.0
zhuldizkuatkizy@cloudshell:~ (elevated-apex-436608-d3)$ mkdir hello_world_app
zhuldizkuatkizy@cloudshell:~ (elevated-apex-436608-d3)$ cd hello_world_app
zhuldizkuatkizy@cloudshell:~/hello_world_app (elevated-apex-436608-d3)$ touch app.py
zhuldizkuatkizy@cloudshell:~/hello_world_app (elevated-apex-436608-d3)$ nano app.py
zhuldizkuatkizy@cloudshell:~/hello_world_app (elevated-apex-436608-d3)$ touch app.yaml
zhuldizkuatkizy@cloudshell:~/hello_world_app (elevated-apex-436608-d3)$ nano app.yaml
zhuldizkuatkizy@cloudshell:~/hello_world_app (elevated-apex-436608-d3)$ gcloud auth login

You are already authenticated with gcloud when running
inside the Cloud Shell and so do not need to run this
command. Do you wish to proceed anyway?

Do you want to continue (Y/n)? n

zhuldizkuatkizy@cloudshell:~/hello_world_app (elevated-apex-436608-d3)$ gcloud config set project elevated-apex-436608-d3
Updated property [core/project].
```

```
zhuldizkuatkizy@cloudshell:~/hello_world_app (elevated-apex-436608-d3)$ gcloud app deploy
You are creating an app for project [elevated-apex-436608-d3].
WARNING: Creating an App Engine application for a project is irreversible and the region
cannot be changed. More information about regions is at
<https://cloud.google.com/appengine/docs/locations>.

Please choose the region where you want your App Engine application located:

[1] asia-east1      (supports standard and flexible)
[2] asia-east2      (supports standard and flexible and search_api)
[3] asia-northeast1 (supports standard and flexible and search_api)
[4] asia-northeast2 (supports standard and flexible and search_api)
[5] asia-northeast3 (supports standard and flexible and search_api)
[6] asia-south1      (supports standard and flexible and search_api)
[7] asia-southeast1 (supports standard and flexible)
[8] asia-southeast2 (supports standard and flexible and search_api)
[9] australia-southeast1 (supports standard and flexible and search_api)
[10] europe-central2 (supports standard and flexible)
[11] europe-west     (supports standard and flexible and search_api)
[12] europe-west2    (supports standard and flexible and search_api)
[13] europe-west3    (supports standard and flexible and search_api)
[14] europe-west6    (supports standard and flexible and search_api)
[15] northamerica-northeast1 (supports standard and flexible and search_api)
[16] southamerica-east1 (supports standard and flexible and search_api)
[17] us-central      (supports standard and flexible and search_api)
[18] us-east1        (supports standard and flexible and search_api)
[19] us-east4        (supports standard and flexible and search_api)
[20] us-west1        (supports standard and flexible)
[21] us-west2        (supports standard and flexible and search_api)
[22] us-west3        (supports standard and flexible and search_api)
[23] us-west4        (supports standard and flexible and search_api)
[24] cancel

Please enter your numeric choice: 8

Creating App Engine application in project [elevated-apex-436608-d3] and region [asia-southeast2]....failed.
ERROR: (gcloud.app.deploy) PERMISSION_DENIED: Read access to project 'elevated-apex-436608-d3' was denied: please check billing account associated and retry.
as zhuldizkuatkizy@gmail.com which is the active account specified by the [core/account] property
zhuldizkuatkizy@cloudshell:~/hello_world_app (elevated-apex-436608-d3)$
```

Error because of billing account

After deploying it should give me URL and in this URL there will be “Hello, world!” message

Exercise 2:

Start your Free Trial with \$300 in credit. Don't worry—you won't be charged if you run out of credits. [Learn more](#)

Google Cloud My Project 65710

Product details

Cloud Functions API

[Google Enterprise API](#)

Manages lightweight user-provided functions executed in response to events.

[ENABLE](#) [TRY THIS API](#)

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API/Service Details [DISABLE API](#)

To use this API, you may need credentials. [CREATE CREDENTIALS](#)

Cloud Functions API

Manages lightweight user-provided functions executed in response to events.

By Google Enterprise API

Service name	Type	Status	Documentation	Explore
cloudfunctions.googleapis.com	Public API	Enabled	LEARN MORE	TRY IN API EXPLORER

[METRICS](#) [QUOTAS & SYSTEM LIMITS](#) [CREDENTIALS](#) [COST](#)

Select Graphs
4 Graphs

1 hour 6 hours 12 hours 1 day 2 days 4 days 7 days 14 days ☒ 30 days

Filters: Versions v1, v2, v2alpha, and v2beta Credentials App Engine default servic... Methods 59 options selected

Google Cloud My Project 65710 cloud functions X Search

Cloud Run functions Functions + CREATE FUNCTION REFRESH

Please make sure to enable billing before creating a function.

Sign up for the free trial to start using Cloud Functions

We will be integrating Cloud Functions into Cloud Run UI in the upcoming months. [LEARN MORE](#)

Filter Filter functions

Environment	Name	Last deployed	Region	Recommendation	Trigger	Runtime	Memory allocated	Executed function	Actions
No rows to display									

There also billing error, but if I didn't have an error:

Then I create the Function:

- Name: helloWorldFunction
- Trigger: HTTP
- Runtime: Node.js 18
- Entry Point: helloWorld

In the **index.js** file:

```
exports.helloWorld = (req, res) => {  
  res.send('Hello, World!');  
};
```

Then I click **Deploy** in the Cloud Console.

After deployment is successful, Google Cloud will provide a URL.

When I open this URL in a browser, I should see "Hello, World!" displayed.

Exercise 3:

```
Use "gcloud config set project [PROJECT_ID]" to change to a different project.  
zhuldizkuatkizy@cloudshell:~ (rock-embassy-437818-m3) $ mkdir docker-app  
zhuldizkuatkizy@cloudshell:~ (rock-embassy-437818-m3) $ cd docker-app  
zhuldizkuatkizy@cloudshell:~/docker-app (rock-embassy-437818-m3) $ touch app.py  
zhuldizkuatkizy@cloudshell:~/docker-app (rock-embassy-437818-m3) $ nano app.py  
zhuldizkuatkizy@cloudshell:~/docker-app (rock-embassy-437818-m3) $ cat app.py  
print("Hello from inside the container!")  
  
zhuldizkuatkizy@cloudshell:~/docker-app (rock-embassy-437818-m3) $ touch Dockerfile  
zhuldizkuatkizy@cloudshell:~/docker-app (rock-embassy-437818-m3) $ nano Dockerfile  
zhuldizkuatkizy@cloudshell:~/docker-app (rock-embassy-437818-m3) $ cat Dockerfile  
FROM python:3.9-slim  
WORKDIR /app  
COPY . /app  
CMD ["python", "app.py"]  
  
zhuldizkuatkizy@cloudshell:~/docker-app (rock-embassy-437818-m3) $ docker build -t hello-world-app .  
[+] Building 8.5s (8/8) FINISHED  
=> [internal] load build definition from Dockerfile  
=> => transferring dockerfile: 110B  
=> [internal] load metadata for docker.io/library/python:3.9-slim  
=> [internal] load .dockerignore  
=> => transferring context: 2B  
=> [1/3] FROM docker.io/library/python:3.9-slim@sha256:49f94609e5a997dc16086a66ac9664591854031d48e375945a9dbf4d1d53abbc  
=> => resolve docker.io/library/python:3.9-slim@sha256:49f94609e5a997dc16086a66ac9664591854031d48e375945a9dbf4d1d53abbc  
=> => sha256:49f94609e5a997dc16086a66ac9664591854031d48e375945a9dbf4d1d53abbc 10.41kB / 10.41kB  
=> => sha256:93ab151da4e5310ea79c4ecf306ece628262b86a4d7a49cc601664f19fe44e36 1.75kB / 1.75kB  
=> => sha256:9d8cb7037cd8e90893e5f430ce4c048a872511e414580c7641675f2dad0a0351 5.20kB / 5.20kB  
=> => sha256:302e3ee498053a7b5332ac79e8efebec16e900289fclcd1c754ce8fa047fcab 29.13MB / 29.13MB  
=> => sha256:4c0965d3919510b506d8856ebc050a96e996c7dae96e4fb420882dbe7e037e67 3.51MB / 3.51MB  
=> => sha256:fdeec85abbad3878f2008f9445f15a19a5a224d1b7e7715ac6b923072333e57 14.74MB / 14.74MB  
=> => sha256:62a08b8dd4f53ad5493dabf2af00ccde91abb3771fb2187040bcf2fe94a7ced7 248B / 248B  
=> => extracting sha256:302e3ee498053a7b5332ac79e8efebec16e900289fclcd1c754ce8fa047fcab  
=> => extracting sha256:4c0965d3919510b506d8856ebc050a96e996c7dae96e4fb420882dbe7e037e67  
=> => extracting sha256:fdeec85abbad3878f2008f9445f15a19a5a224d1b7e7715ac6b923072333e57  
=> => extracting sha256:62a08b8dd4f53ad5493dabf2af00ccde91abb3771fb2187040bcf2fe94a7ced7  
=> [internal] load build context  
=> => transferring context: 188B  
=> [2/3] WORKDIR /app  
=> [3/3] COPY . /app  
=> exporting to image  
=> => exporting layers  
=> => writing image sha256:5d56f4f9f827b38426c1153ba8747a36fd816fe4d98935474cla869c0b9ac6bb  
=> => naming to docker.io/library/hello-world-app  
zhuldizkuatkizy@cloudshell:~/docker-app (rock-embassy-437818-m3) $ docker images  
REPOSITORY TAG IMAGE ID CREATED SIZE  
hello-world-app latest 5d56f4f9f827 34 seconds ago 125MB  
zhuldizkuatkizy@cloudshell:~/docker-app (rock-embassy-437818-m3) $ docker run --rm hello-world-app  
Hello from inside the container!  
zhuldizkuatkizy@cloudshell:~/docker-app (rock-embassy-437818-m3) $
```